

8(a)(i)	A photon is a quantum of electromagnetic radiation with a discrete amount of energy $E = hf$, where h is Planck's constant and f is the frequency of the radiation.
8(a)(ii)	Coherent light waves are light waves that have a constant phase difference.
8(b)(i)	Using $\sin \theta = \frac{\lambda}{b}$, $\sin \theta = \frac{5.90 \times 10^{-7}}{0.15 \times 10^{-3}} = 3.933 \times 10^{-3}$ $\theta = 3.933 \times 10^{-3} \text{ rad}$ Along the screen, the distance from the central axis to the first minimum on either side is x.

No.	Solution
	$\tan \theta = \frac{x}{3.2}$ $x = 3.2 \tan(3.933 \times 10^{-3})$ $= 0.0126 \text{ m}$ Width of central bright maximum $= 2x$ $= 2 \times 0.0126$ $= 0.0252 \text{ m}$
8(b)(ii)	The Rayleigh criterion states that the images of two sources are just resolvable when the central maximum of one diffraction pattern image falls on the first minimum of the other diffraction pattern image.
8(b)(iii)	The separation between the two slits is 2.3 mm. This is very much smaller than the separation of 12.6 mm between the centre of the diffraction pattern of one to the first minimum of the other. Hence, the two diffraction pattern images are not seen on the screen as being separate.
8(b)(iv)	Using $x = \frac{\lambda D}{a}$, width of a fringe = fringe separation, x $= \frac{(5.90 \times 10^{-7})(3.2)}{2.3 \times 10^{-3}}$ $= 8.2 \times 10^{-4} \text{ m}$
8(b)(v)	Number of fringes $= \frac{0.0252}{8.2 \times 10^{-4}}$ $= 30$
8(c)(i)	Using $d \sin \theta = n\lambda$, $\sin \theta = \frac{n\lambda}{d}$ $\therefore \sin \theta \leq 1, \frac{n\lambda}{d} \leq 1,$ $n \leq \frac{d}{\lambda} = \frac{1 \times 10^{-3}}{5.90 \times 10^{-7}} = 2.3$ So, maximum order of fringe observable is $n = 2$. $n = 1, \theta = \sin^{-1} \left(\frac{5.90 \times 10^{-7}}{\frac{1 \times 10^{-3}}{750}} \right)$ $= 26.3^\circ$

No.	Solution
	$n = 2, \theta = \sin^{-1} \left(\frac{\frac{2 \times 5.90 \times 10^{-7}}{1 \times 10^{-3}}}{750} \right)$ $= 62.3^\circ$ Hence, the angles at which maxima are seen are 0°, 26.3° and 62.3°.
8(c)(ii)	If the incident beam makes an angle with the grating normal, the expression $d \sin \theta = n\lambda$ is no longer true. Instead of the fringe separation increasing with increasing order on both sides of the central maximum, the angles at which the increasing order of the fringes make with the central maximum will increase with increasing order on one side and decrease with increasing order on the other side of the central maximum. Note: Question did not ask for explanation. So no need to explain.
8(c)	Radioactive decay is a spontaneous and random process whereby an unstable